

## SAFETY DATA SHEET

Creation Date 22-Sep-2009

Revision Date 29-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Tetranitro blue tetrazolium chloride

**Cat No. :** J61080

**CAS No** 1184-43-6  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Carcinogenicity

Category 1A

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use

Do not handle until all safety precautions have been read and understood

Use personal protective equipment as required

**Response**

IF exposed or concerned: Get medical attention/advice

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

WARNING. Reproductive Harm - <https://www.p65warnings.ca.gov/>.

### 3. Composition/Information on Ingredients

| Component                                                                                                | CAS No    | Weight % |
|----------------------------------------------------------------------------------------------------------|-----------|----------|
| 2H-Tetrazolium,<br>3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride | 1184-43-6 | 96-97    |
| Ethyl alcohol                                                                                            | 64-17-5   | 3-4      |

### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                          |
|---------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b> | No information available |
| <b>Flash Point</b>                    | No information available |
| <b>Method -</b>                       | No information available |

|                                  |                          |
|----------------------------------|--------------------------|
| Autoignition Temperature         | Not applicable           |
| Explosion Limits                 |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |
| Sensitivity to Mechanical Impact | No information available |
| Sensitivity to Static Discharge  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Carbon dioxide (CO<sub>2</sub>). Carbon monoxide (CO). Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

Health  
2

Flammability  
3

Instability  
1

Physical hazards  
N/A

**6. Accidental release measures**

|                           |                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------|
| Personal Precautions      | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. |
| Environmental Precautions | Do not flush into surface water or sanitary sewer system.                                         |

|                                      |                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Methods for Containment and Clean Up | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|

**7. Handling and storage**

|          |                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|          |                                                                          |
|----------|--------------------------------------------------------------------------|
| Storage. | Keep containers tightly closed in a dry, cool and well-ventilated place. |
|----------|--------------------------------------------------------------------------|

**8. Exposure controls / personal protection****Exposure Guidelines**

| Component     | ACGIH TLV      | OSHA PEL                                                                                                         | NIOSH                                                          | Mexico OEL (TWA) |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------|
| Ethyl alcohol | STEL: 1000 ppm | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> | IDLH: 3300 ppm<br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> | STEL: 1000 ppm   |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

|                      |                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Engineering Measures | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                                 |
|-----------------------------------------------|---------------------------------|
| <b>Physical State</b>                         | Powder Solid                    |
| <b>Appearance</b>                             | Yellow - Brown                  |
| <b>Odor</b>                                   | No information available        |
| <b>Odor Threshold</b>                         | No information available        |
| <b>pH</b>                                     | No information available        |
| <b>Melting Point/Range</b>                    | 161 - 166 °C / 321.8 - 330.8 °F |
| <b>Boiling Point/Range</b>                    | No information available        |
| <b>Flash Point</b>                            | No information available        |
| <b>Evaporation Rate</b>                       | Not applicable                  |
| <b>Flammability (solid,gas)</b>               | No information available        |
| <b>Flammability or explosive limits</b>       |                                 |
| Upper                                         | No data available               |
| Lower                                         | No data available               |
| <b>Vapor Pressure</b>                         | No information available        |
| <b>Vapor Density</b>                          | Not applicable                  |
| <b>Specific Gravity</b>                       | No information available        |
| <b>Solubility</b>                             | No information available        |
| <b>Partition coefficient; n-octanol/water</b> | No data available               |
| <b>Autoignition Temperature</b>               | Not applicable                  |
| <b>Decomposition Temperature</b>              | No information available        |
| <b>Viscosity</b>                              | Not applicable                  |
| <b>Molecular Formula</b>                      | C40H28Cl2N12O10.xC2H5OH         |
| <b>Molecular Weight</b>                       | 907.64                          |

## 10. Stability and reactivity

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                            |
| <b>Stability</b>                        | Stable under normal conditions.                                                                       |
| <b>Conditions to Avoid</b>              | Incompatible products.                                                                                |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                                               |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx), Carbon dioxide (CO <sub>2</sub> ), Carbon monoxide (CO), Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                         |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

### Component Information

| Component     | LD50 Oral                            | LD50 Dermal | LC50 Inhalation                            |
|---------------|--------------------------------------|-------------|--------------------------------------------|
| Ethyl alcohol | LD50 = 10470 mg/kg<br>OECD 401 (Rat) | Not listed  | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat) |

|  |                      |  |                     |
|--|----------------------|--|---------------------|
|  | 3450 mg/kg ( Mouse ) |  | 20000 ppm/10H (rat) |
|--|----------------------|--|---------------------|

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** Possible cancer hazard. May cause cancer based on animal data. Ethanol has been shown to be carcinogenic in long-term studies only when consumed and abused as an alcoholic beverage. The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                                                             | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------------------------------------------------------------------------------|-----------|------------|------------|------------|------------|------------|
| 2H-Tetrazolium, 3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride | 1184-43-6 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Ethyl alcohol                                                                                         | 64-17-5   | Not listed | Known      | A3         | Not listed | A3         |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*NTP: (National Toxicity Program)*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component     | Freshwater Algae                              | Freshwater Fish                                                  | Microtox                                                                                                  | Water Flea                                    |
|---------------|-----------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Ethyl alcohol | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | Photobacterium<br>phosphoreum: EC50 = 34634<br>mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 = 35470 | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h |

|  |  |  |            |  |
|--|--|--|------------|--|
|  |  |  | mg/L/5 min |  |
|--|--|--|------------|--|

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

| Component     | log Pow |
|---------------|---------|
| Ethyl alcohol | -0.32   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN2926  
 Proper Shipping Name FLAMMABLE SOLID, TOXIC, ORGANIC, N.O.S.  
 Hazard Class 4.1  
 Subsidiary Hazard Class 6.1  
 Packing Group III

#### TDG

UN-No UN2926  
 Proper Shipping Name FLAMMABLE SOLID, TOXIC, ORGANIC, N.O.S.  
 Hazard Class 4.1  
 Subsidiary Hazard Class 6.1  
 Packing Group III

**IATA** Not regulated

**IMDG/IMO** Not regulated

### 15. Regulatory information

#### United States of America Inventory

| Component                                                                                             | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------------------------------------------------------------------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| 2H-Tetrazolium, 3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride | 1184-43-6 | X    | INACTIVE                                      | -                           |
| Ethyl alcohol                                                                                         | 64-17-5   | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

#### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component       | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-----------------|-----------|-----|-----|-----------|-------|------|------|------|-------|------|
| 2H-Tetrazolium, | 1184-43-6 | -   | X   | 214-665-1 | -     | -    |      | -    | -     | -    |

|                                                                                       |         |   |   |           |   |   |   |   |   |          |
|---------------------------------------------------------------------------------------|---------|---|---|-----------|---|---|---|---|---|----------|
| 3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride |         |   |   |           |   |   |   |   |   |          |
| Ethyl alcohol                                                                         | 64-17-5 | X | - | 200-578-6 | X | X | X | X | X | KE-13217 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product contains the following Proposition 65 chemicals.

| Component     | CAS No  | California Prop. 65                                  | Prop 65 NSRL | Category                    |
|---------------|---------|------------------------------------------------------|--------------|-----------------------------|
| Ethyl alcohol | 64-17-5 | Development (alcoholic beverages only)<br>Carcinogen | -            | Developmental<br>Carcinogen |

### **U.S. State Right-to-Know Regulations**

| Component     | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------|---------------|------------|--------------|----------|--------------|
| Ethyl alcohol | X             | X          | X            | X        | X            |

### **U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component | CAS No | REACH (1907/2006) -<br>Annex XIV - Substances | REACH (1907/2006) -<br>Annex XVII - Restrictions | REACH Regulation (EC<br>1907/2006) article 59 - |
|-----------|--------|-----------------------------------------------|--------------------------------------------------|-------------------------------------------------|
|-----------|--------|-----------------------------------------------|--------------------------------------------------|-------------------------------------------------|

|                                                                                                       |           | Subject to Authorization | on Certain Dangerous Substances | Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------------------------------------------------------------------------------|-----------|--------------------------|---------------------------------|----------------------------------------------------------|
| 2H-Tetrazolium, 3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride | 1184-43-6 | -                        | -                               | -                                                        |
| Ethyl alcohol                                                                                         | 64-17-5   | -                        | -                               | -                                                        |

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                                                                             | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------------------------------------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2H-Tetrazolium, 3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride | 1184-43-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Ethyl alcohol                                                                                         | 64-17-5   | Listed         | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component                                                                                             | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2H-Tetrazolium, 3,3'-(3,3'-dimethoxy[1,1'-biphenyl]-4,4'-diyl)bis[2,5-bis(4-nitrophenyl)-, dichloride | 1184-43-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Ethyl alcohol                                                                                         | 64-17-5   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

22-Sep-2009

## Revision Date

29-Mar-2024

## Print Date

29-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS